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Chapter 1: Vehicle cruise control

1.1 Model and parameters

The objective of the cruise-control system is to keep the longitudinal vehicle speed close to a selected reference speed even when the road slope, rolling resistance, aerodynamic drag or vehicle mass change. The controller measures the speed error and adjusts the throttle command. In this study a PI controller is used because the integral term removes steady-state error caused by constant disturbances such as a constant road grade.

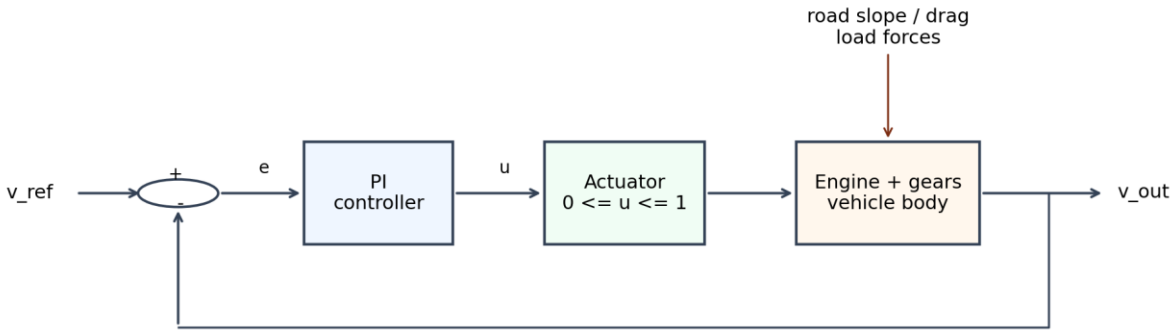


Figure 1. Schematic feedback loop for the cruise-control problem.

The vehicle model is obtained from the longitudinal force balance. The variable v is the vehicle speed, u is the normalized throttle position, and Fd denotes the sum of disturbance forces.

$$m\,dv/dt = F - Fd$$

The engine torque at full throttle is approximated by a parabolic curve in the engine angular speed. With the gear coefficient alpha_n, the engine speed is omega = alpha_n v and the driving force is alpha_n u T(alpha_n v).

$$T(\text{omega}) = Tm\,[1 - \text{beta}\,(\text{omega}/\text{omega_m} - 1)^2]$$

$$dv/dt = (\text{alpha_n}\,u\,Tm/m)[1 - \text{beta}(\text{alpha_n}\,v/\text{omega_m} - 1)^2] - g\,cr\,\text{sign}(v) - (\text{rho}\,cd\,A/(2m))v^2 - g\,\sin(\text{theta})$$

Parameter	Value	Parameter	Value
Tm	190 N m	omega_m	420 rad/s
beta	0.4	rho	1.3 kg/m^3
cd	0.32	A	2.4 m^2
cr	0.01	g	9.81 m/s^2
kp	0.5	ki	0.1

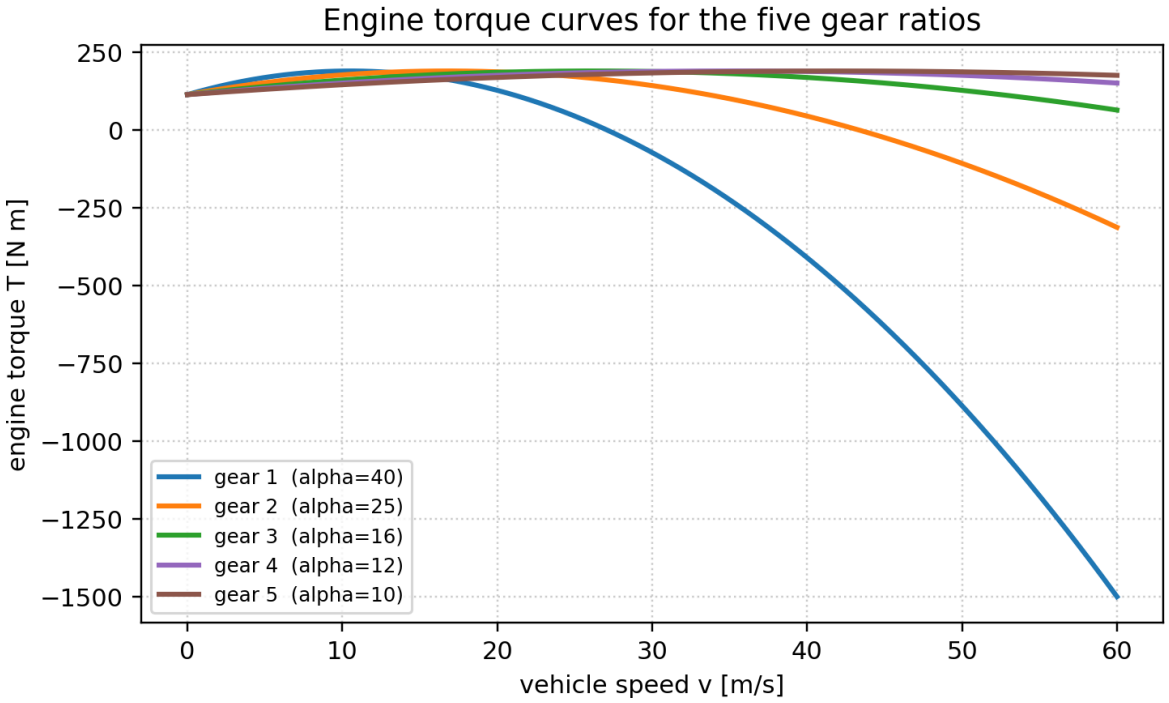


Figure 2. Engine torque as a function of vehicle speed for the five gear ratios.

The torque curves show that higher gears move the useful torque region toward higher vehicle speed. This explains why the same hill disturbance requires different throttle corrections depending on the gear ratio.

1.2 Linearisation around an operating point

For local robustness analysis, the nonlinear model is linearised around an equilibrium pair (u_e, v_e). At equilibrium the acceleration is zero and the throttle value is obtained by balancing the engine force with drag, rolling resistance and gravity.

$$u_e = [m\ g\ cr\ sign(v_r) + 0.5\ rho\ cd\ A\ v_r^2 + m\ g\ sin(theta)]\ /\ [alpha_n\ Tm\ (1 - beta(alpha_n\ v_r/omega_m - 1)^2)]$$

For gear 5, mass m = 1000 kg, theta = 0 and v_r = 20 m/s, the equilibrium throttle is u_e = 0.1760. This is the initial condition used for the integrator in the nonlinear simulations.

$$d(v - v_e)/dt = A(v - v_e) + B(u - u_e),\ y = v - v_e$$

$$A = (alpha_n\ u_e\ Tm/m)[-2\ beta(alpha_n\ v_e/omega_m - 1)(alpha_n/omega_m)] - (rho\ cd\ A_area\ v_e/m)$$

$$B = (alpha_n\ Tm/m)[1 - beta(alpha_n\ v_e/omega_m - 1)^2]$$

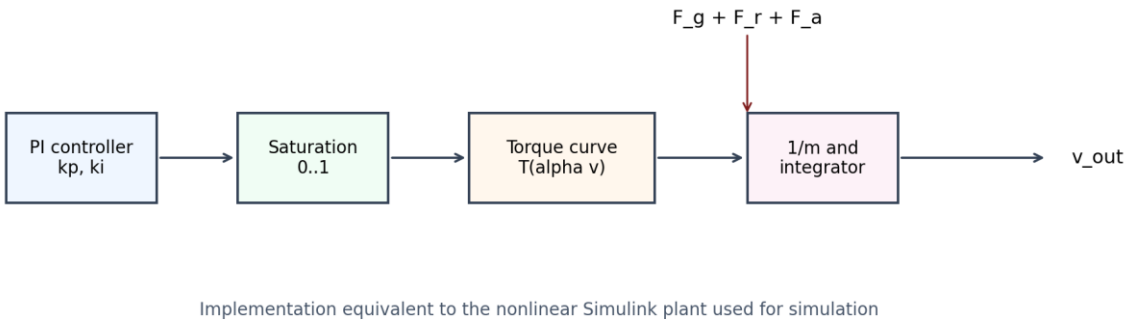


Figure 3. Block representation of the nonlinear model used for the simulations.

1.3 Hill disturbance simulations

The hill is modelled as a ramp in the road slope. The angle starts from 0 deg at t = 5 s and reaches 4 deg at t = 6 s. All simulations begin from the equilibrium condition for v_r = 20 m/s.

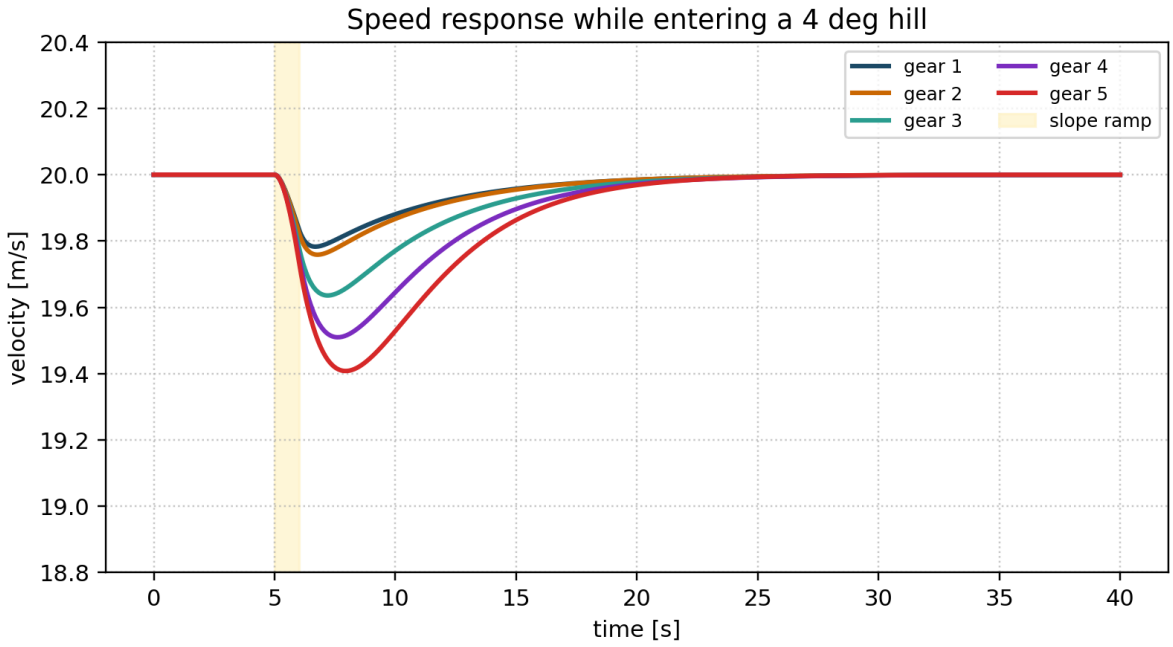


Figure 4. Vehicle velocity for five gears while entering a 4 deg hill.

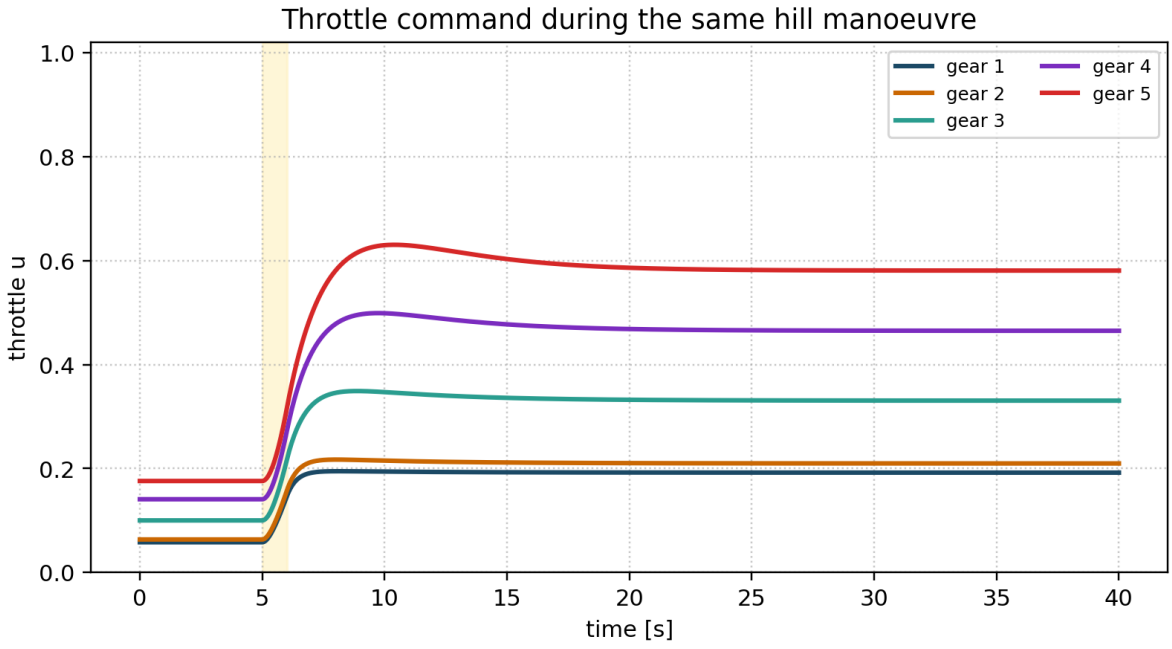


Figure 5. Throttle command for five gears during the same hill perturbation.

The largest speed deviation is obtained in the highest gear because the available acceleration is smaller around this operating condition. The PI action increases the throttle until the additional gravitational load is compensated. For gear 5 the final equilibrium throttle for a 4 deg hill is about 0.5806.

1.4 Anti-windup correction

When the required throttle is larger than the physical limit u = 1, the integrator of a PI controller may continue to increase even though the actuator is saturated. This is the windup problem. A back-calculation anti-windup loop is therefore added.

$$C(s) = kp(1 + 1/(s\ Ti)), \quad Ti = kp/ki, \quad kt = 1/Ti = ki/kp = 0.2$$

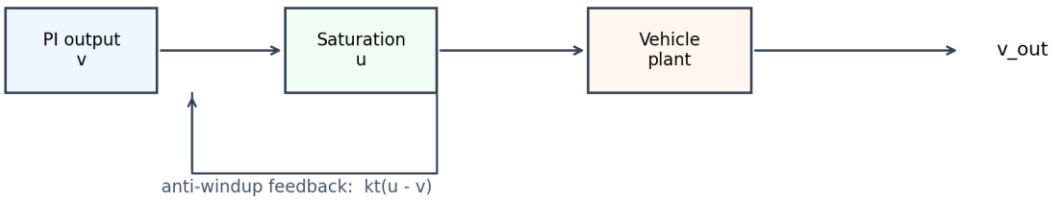


Figure 6. Anti-windup back-calculation structure.

A stronger hill perturbation is used here: the slope increases from 0 deg to 8 deg. This case pushes the actuator near its upper limit; the corresponding final equilibrium throttle is approximately 0.9832 for gear 5.

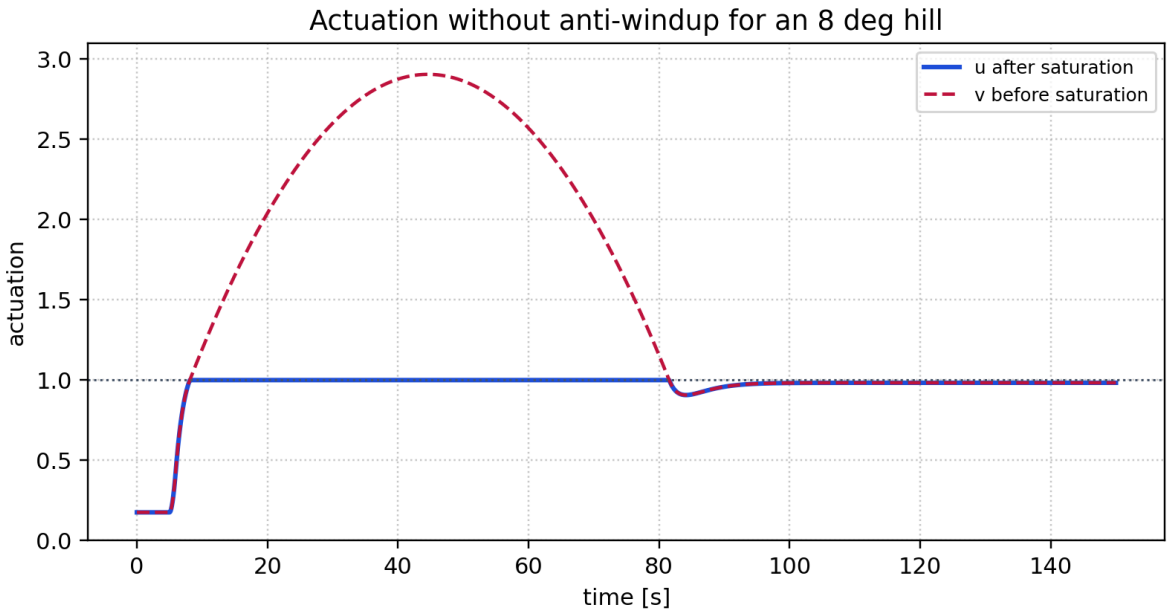


Figure 7. Actuator signal without anti-windup correction.

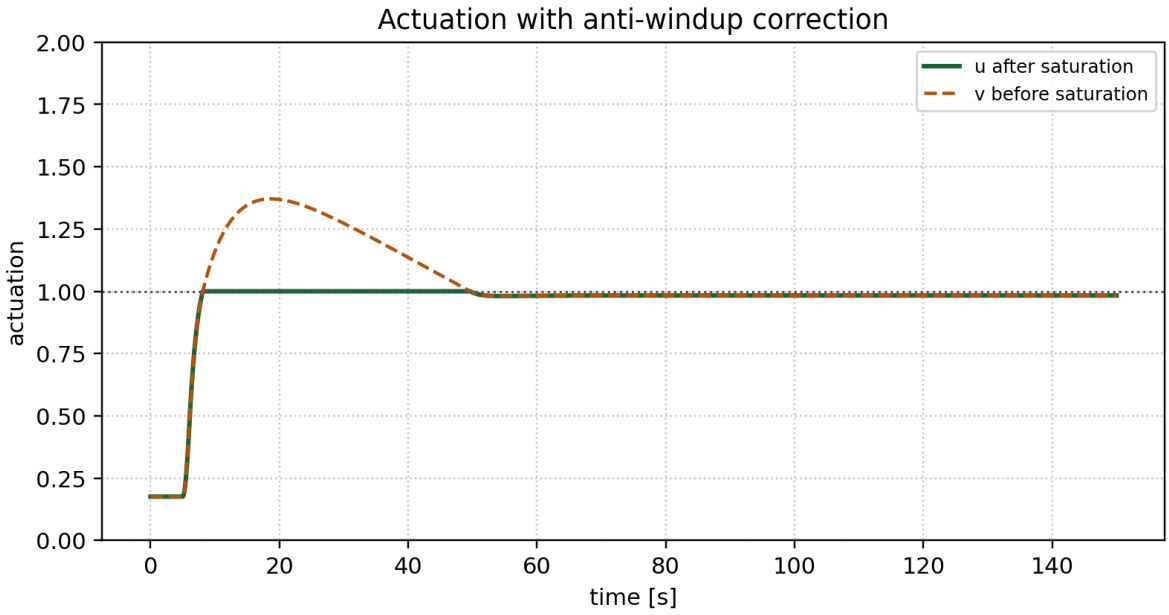


Figure 8. Actuator signal with anti-windup correction.

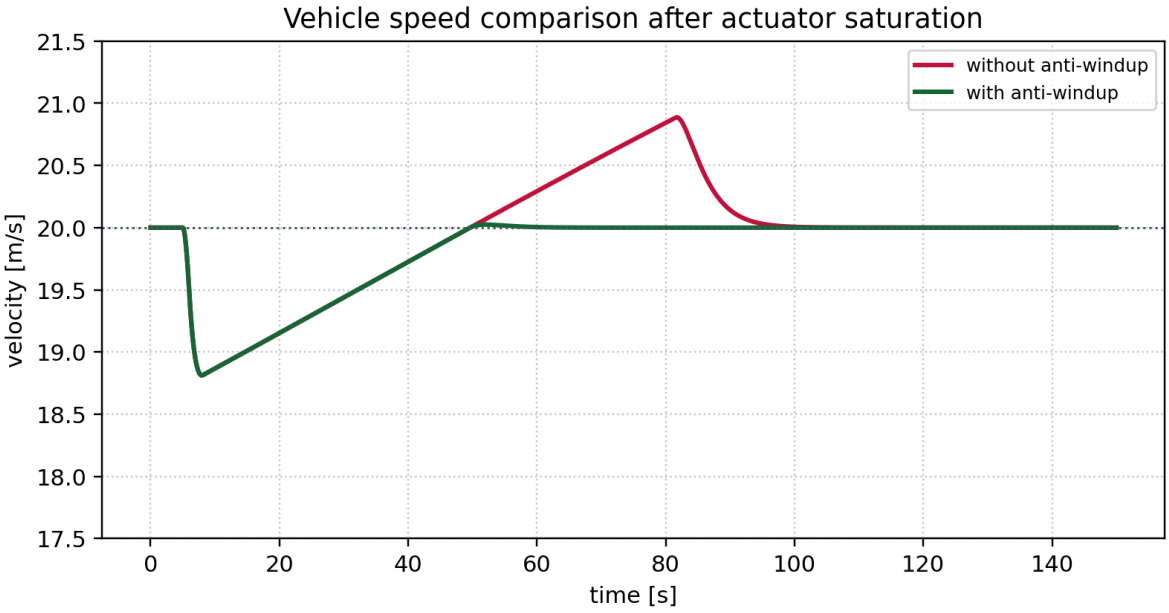


Figure 9. Speed response comparison with and without anti-windup.

The anti-windup term prevents the internal controller output from growing excessively during saturation. Consequently, when the vehicle reaches the new operating region, the speed returns to the reference with less overshoot and a shorter recovery time.

1.5 Robustness evaluation

To test robustness against plant-parameter changes, the same PI controller is used for three masses: 1000 kg, 2000 kg and 3000 kg. The reference speed changes from 25 m/s to 30 m/s. Anti-windup is kept active because the reference step temporarily saturates the throttle.

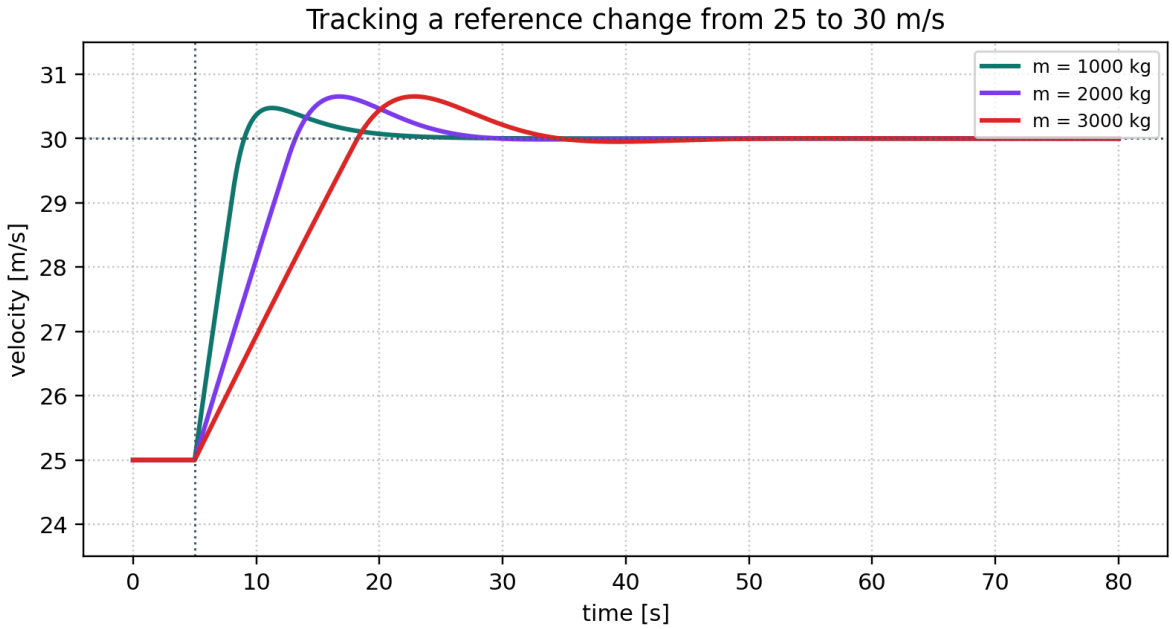


Figure 10. Velocity tracking for three different vehicle masses.

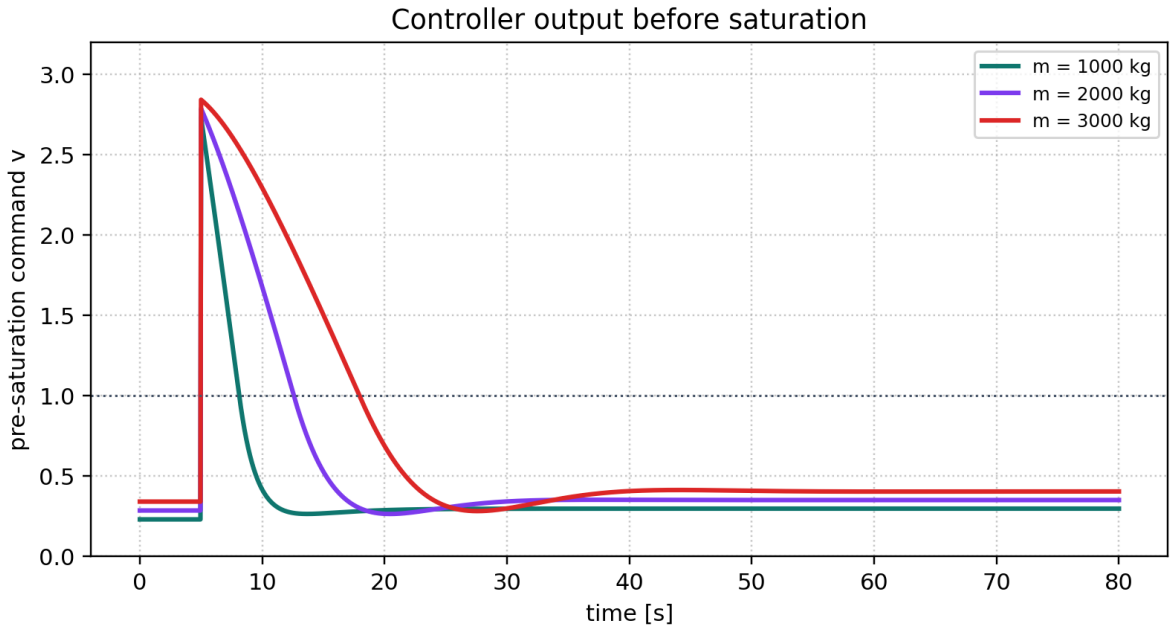


Figure 11. Controller output before saturation for the three masses.

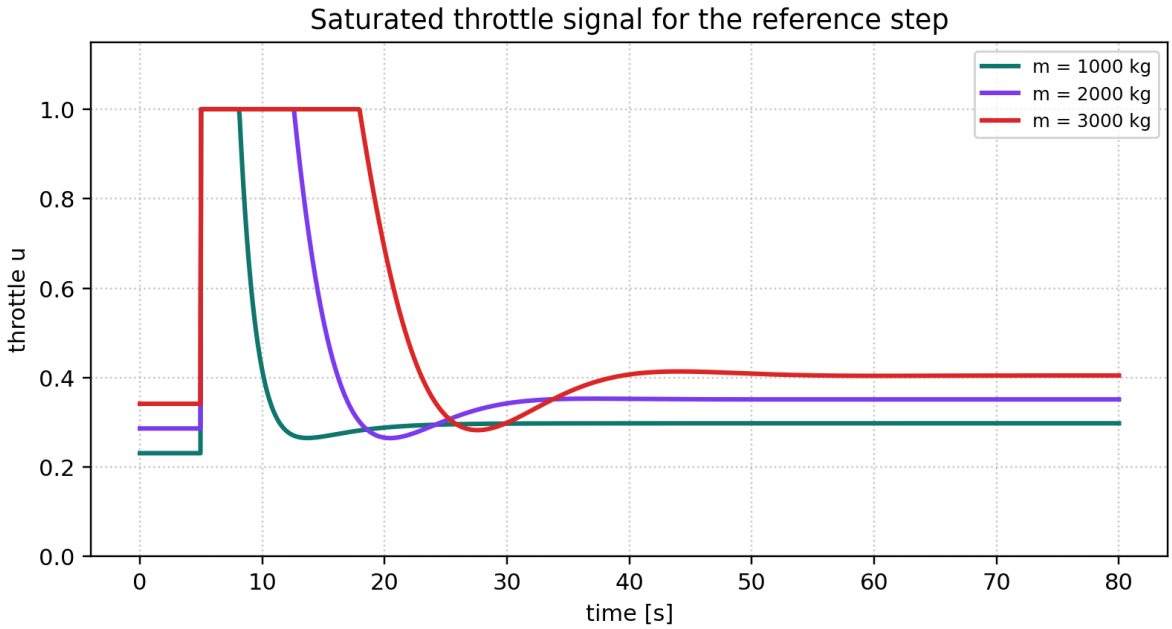


Figure 12. Saturated throttle command for the reference step.

The lighter vehicle reaches the new speed more quickly, while the heavier vehicle remains saturated for a longer time. Nevertheless, all three cases converge to the desired speed, which confirms the usefulness of the integral action for low-frequency disturbances and mass variations.

A second robustness test combines the reference step with a slope change from 0 deg to 2 deg between t = 5 s and t = 6 s. This creates simultaneous reference and load disturbances.

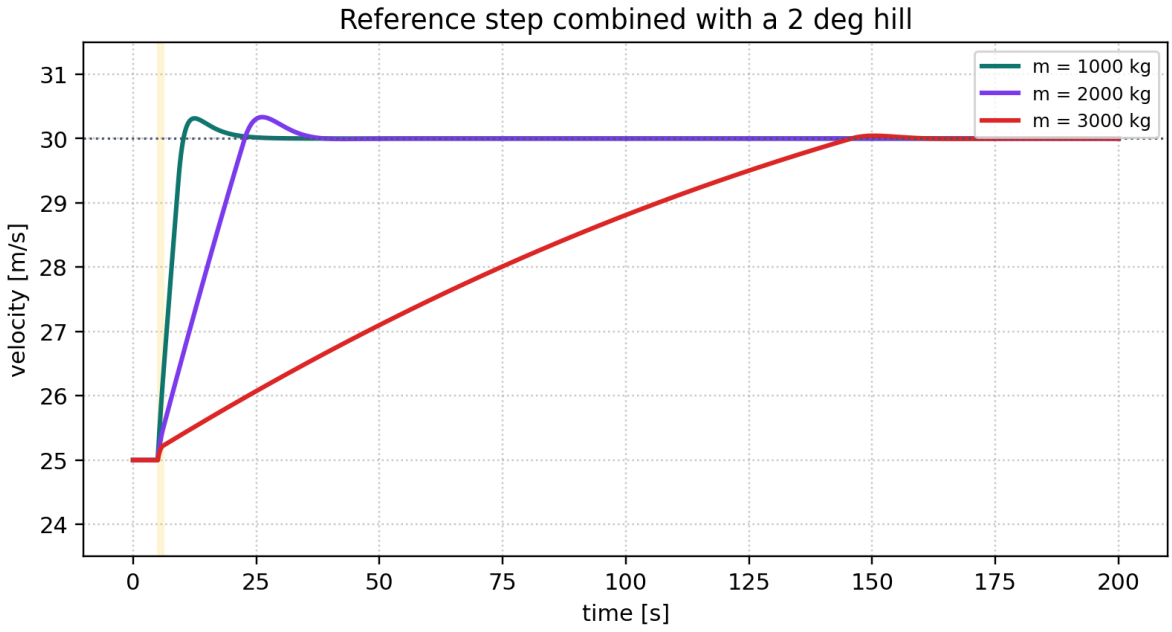


Figure 13. Velocity response under simultaneous reference and slope disturbances.

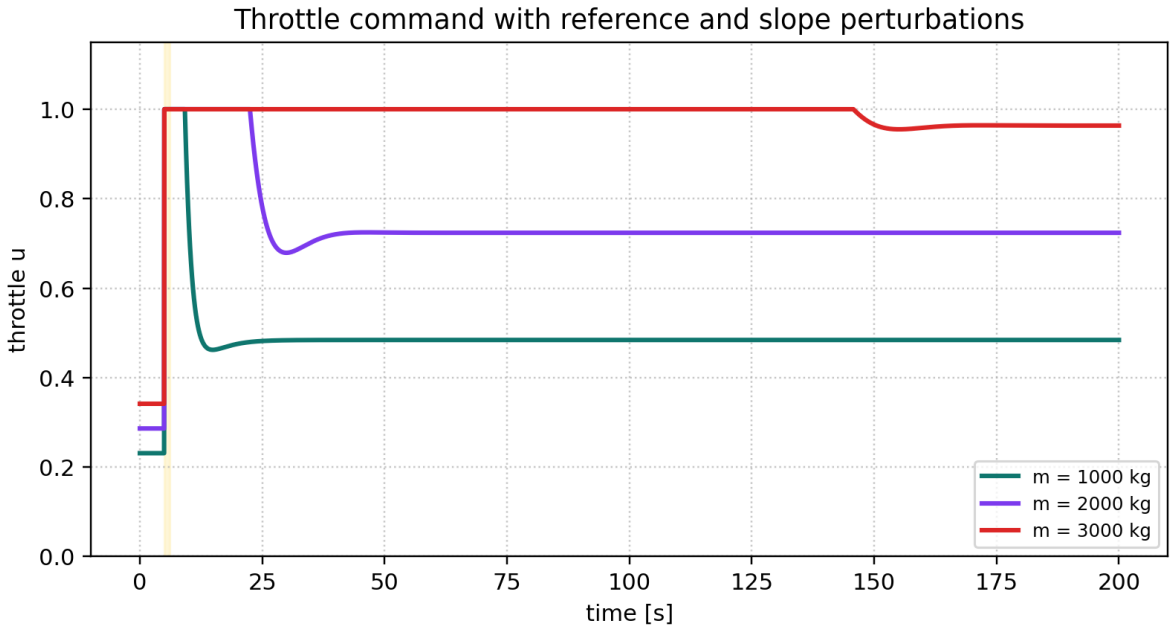


Figure 14. Throttle command under simultaneous reference and slope disturbances.

The controller again rejects the low-frequency disturbance and reaches the new reference. The settling time increases with vehicle mass because the same maximum throttle produces less acceleration for heavier vehicles.

1.6 Gang-of-four transfer functions

The local transfer-function analysis is performed from the linearised model. To keep the numerical values consistent with the MATLAB robustness script, the operating point used here is gear 5, m = 3000 kg, v_e = 25 m/s and theta = 0. This gives:

$$A = -0.006653, \quad B = 0.591829, \quad u_e = 0.341484$$

$$P(s) = B/(s - A), \quad C(s) = kp + ki/s$$

$$S(s) = 1/(1 + P(s)C(s)) = [s^2 - A\ s]/[s^2 + (B\ kp - A)s + B\ ki]$$

$$P(s)S(s) = [B\ s]/[s^2 + (B\ kp - A)s + B\ ki]$$

$$C(s)S(s) = [kp\ s^2 + (ki - A\ kp)s - A\ ki]/[s^2 + (B\ kp - A)s + B\ ki]$$

$$T(s) = P(s)C(s)/(1 + P(s)C(s)) = [B\ kp\ s + B\ ki]/[s^2 + (B\ kp - A)s + B\ ki]$$

All denominator coefficients are positive and the roots are in the left half-plane. Therefore, the four transfer functions are BIBO stable. The sensitivity has a zero at the origin, so constant disturbances are rejected at steady state.

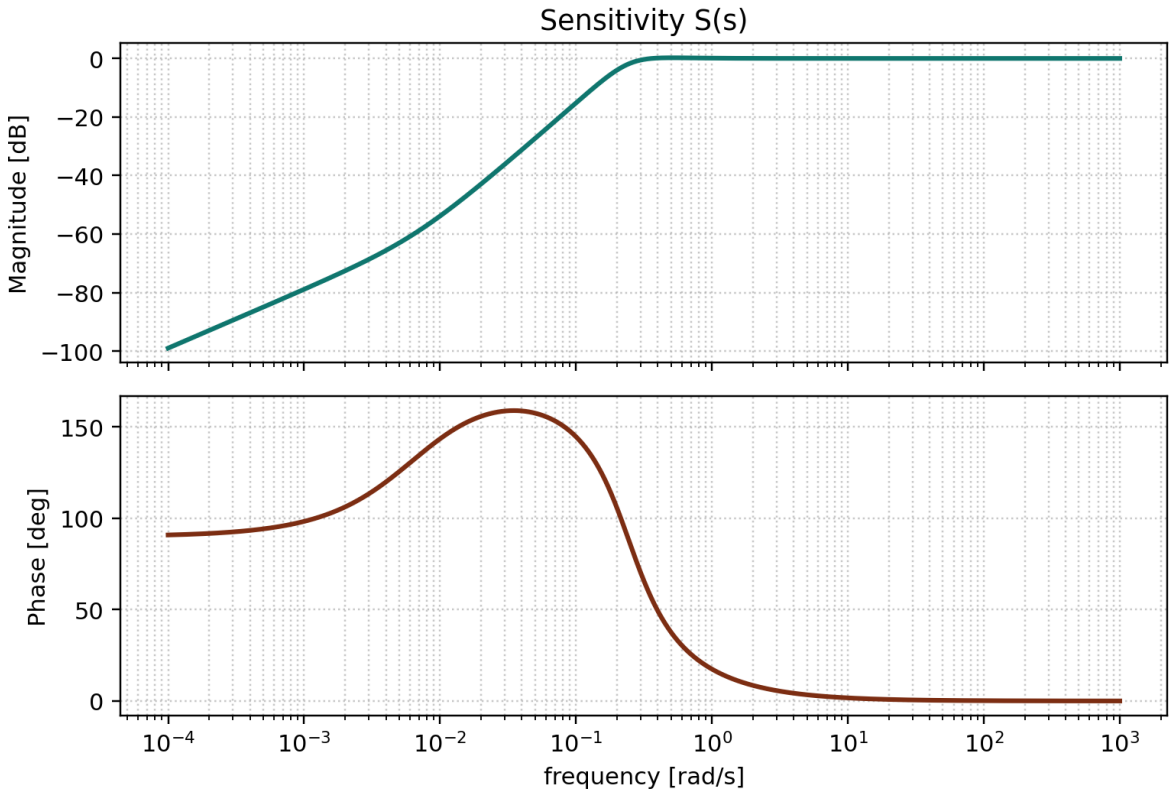


Figure 15. Bode diagram of the sensitivity transfer function S(s).

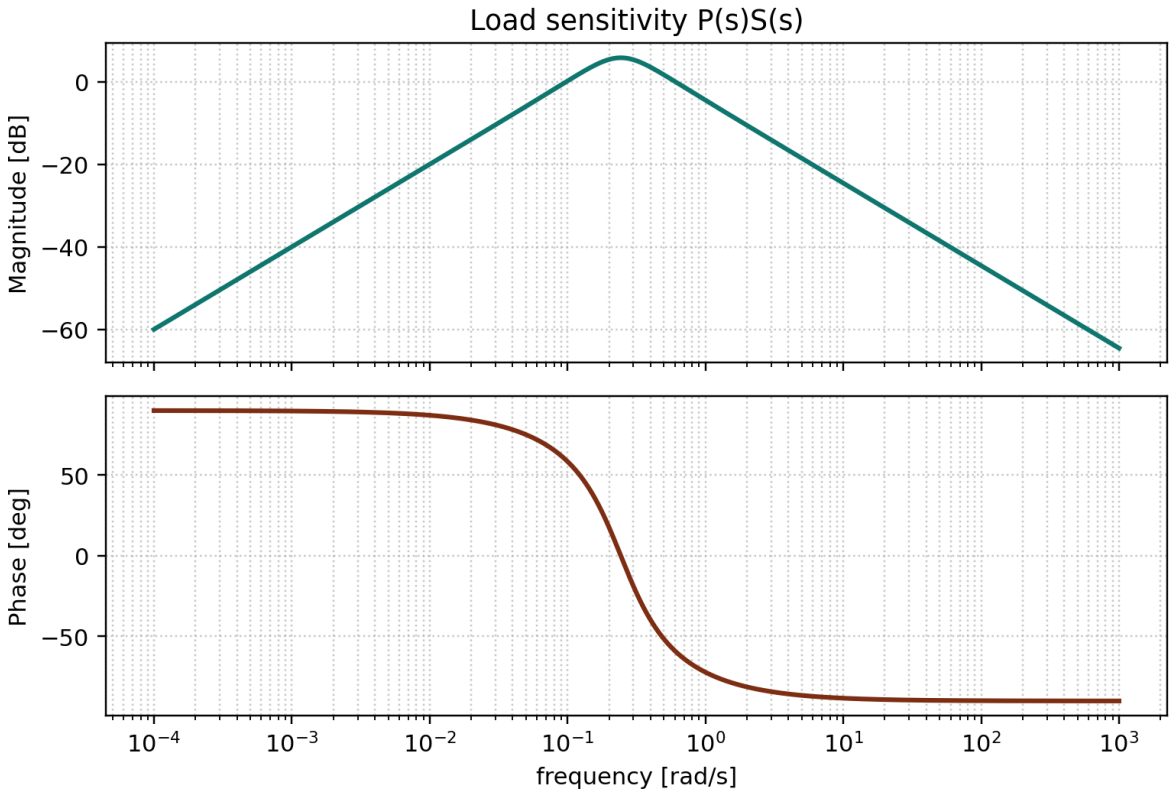


Figure 16. Bode diagram of load sensitivity P(s)S(s).

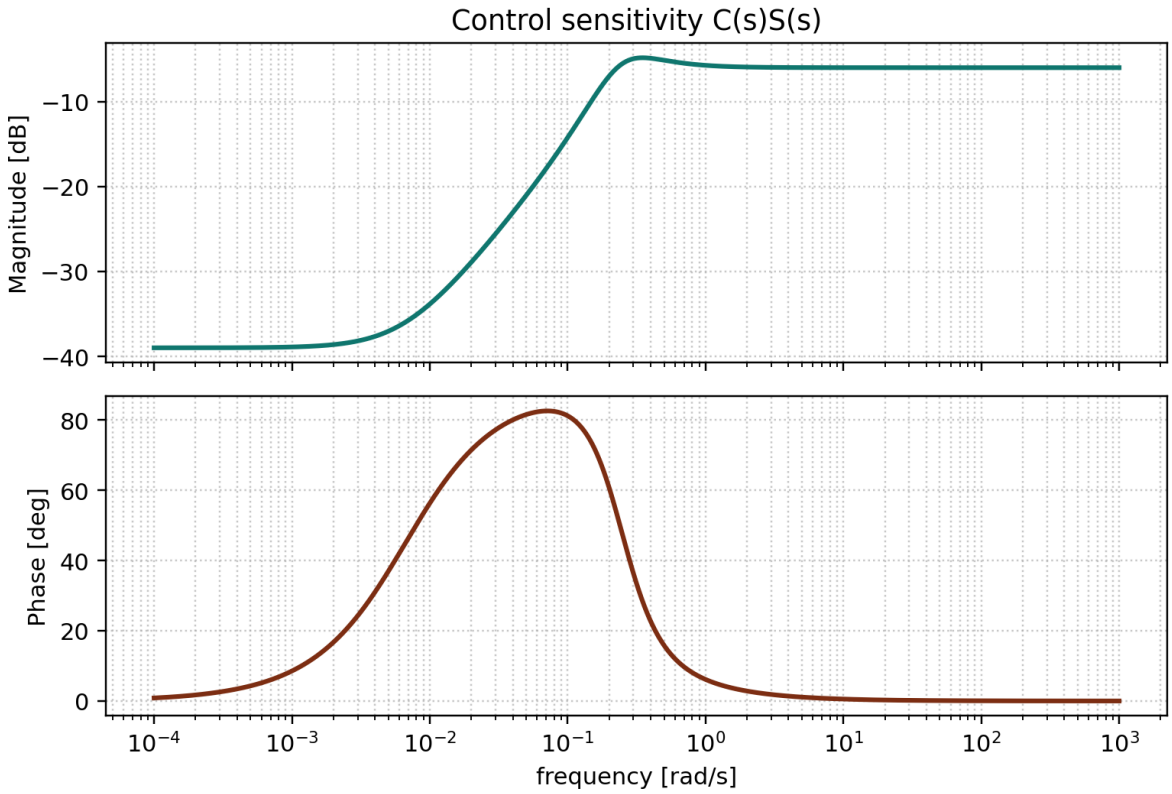


Figure 17. Bode diagram of control sensitivity C(s)S(s).

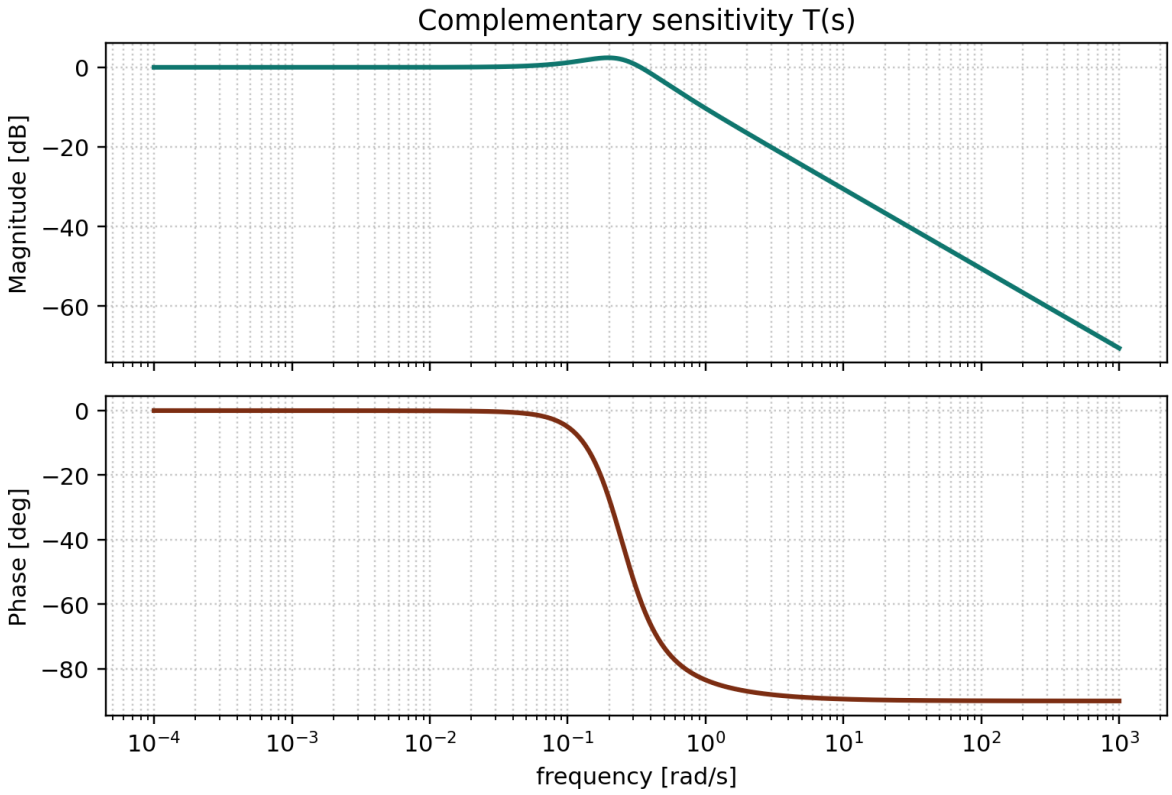


Figure 18. Bode diagram of complementary sensitivity T(s).

Transfer function	Peak magnitude	Peak frequency [rad/s]
Sensitivity S(s)	1.0268	0.5098
Load sensitivity P(s)S(s)	1.9560	0.2429
Control sensitivity C(s)S(s)	0.5719	0.3490
Complementary sensitivity T(s)	1.3188	0.1961

The maximum value of $|S(jw)|$ is 1.0268, giving a stability margin approximately equal to $1/\max|S| = 0.9739$. The complementary sensitivity is close to 1 at low frequency, which is consistent with good tracking of slowly varying reference commands.